Chapter 2. Describing Contingency Tables

Proportions, relative risk, odds ratios, partial association

Contents

2.1 Probability structure for contingency tables **2**
- 2.1.1 Contingency tables and their distributions . . . 2
- 2.1.2 Sensitivity and specificity . . . 2
- 2.1.3 Independence of categorical variables . . . 3
- 2.1.4 Poisson, binomial, and multinomial sampling . . . 3
- 2.1.5 Seat belt example . . . 4
- 2.1.6 Types of studies . . . 4

2.2 Comparing two proportions **5**
- 2.2.1 Difference of proportions . . . 5
- 2.2.2 Relative risk . . . 5
- 2.2.3 Odds ratio . . . 5
- 2.2.4 Properties of the odds ratio . . . 5
- 2.2.5 Aspirin and heart attacks revisited . . . 6
- 2.2.6 Case–control studies and the odds ratio . . . 6
- 2.2.7 Relationship between odds ratio and relative risk . . . 7

2.3 Partial association in stratified 2×2 tables **7**
- 2.3.1 Partial tables . . . 7
- 2.3.2 Death penalty example . . . 7
- 2.3.3 Conditional and marginal odds ratios . . . 9
- 2.3.4 Marginal versus conditional independence . . . 9
- 2.3.5 Homogeneous association . . . 10

2.4 Extensions for $I \times J$ tables **11**
- 2.4.1 Odds ratios in $I \times J$ tables . . . 11
- 2.4.2 Summary measures of association . . . 12
- 2.4.3 Ordinal trends: concordant and discordant pairs . . . 12
- 2.4.4 Ordinal measure of association: gamma . . . 13
- 2.4.5 Gamma for job satisfaction example . . . 14

2.1 Probability structure for contingency tables

Motivation. The relationship between two categorical variables is entirely contained in their joint distribution, from which the marginal and conditional distributions follow. How the data were sampled determines which of these distributions the observed table can estimate.

2.1.1 Contingency tables and their distributions

Let X have I categories and Y have J categories. A rectangular table with I rows and J columns whose IJ cells hold frequency counts is a *contingency table* (Karl Pearson, 1904), or $I \times J$ cross-classification table.

Definition (Joint, marginal and conditional distributions)

(a) The *joint distribution* is $\{\pi_{ij}\}$, $\pi_{ij} = P(X = i, Y = j)$.
(b) The *marginal distributions* are $\pi_{i+} = \sum_j \pi_{ij}$ and $\pi_{+j} = \sum_i \pi_{ij}$, with $\sum_i \pi_{i+} = \sum_j \pi_{+j} = \sum_i \sum_j \pi_{ij} = 1$ (a "+" denotes summation over that index).
(c) When Y is a response, the *conditional distribution* of Y in row i is $\pi_{j|i} = P(Y = j \mid X = i) = \pi_{ij}/\pi_{i+}$, with $\sum_j \pi_{j|i} = 1$.

If X is fixed by design (not random), only the conditional distributions are meaningful. A principal aim of many studies is to compare the conditional distributions of Y across levels of X.

Example 2.1 (Aspirin and heart attacks)

The Physicians' Health Study was a five-year randomized, blind trial of whether regular aspirin intake reduces cardiovascular mortality. Every other day physicians took aspirin or a placebo.

	Myocardial infarction			
Group	Fatal	Nonfatal	No attack	Total
Placebo	18	171	10,845	11,034
Aspirin	5	99	10,933	11,037

Of 11,034 placebo takers 18 had fatal attacks, against 5 of 11,037 aspirin takers.

2.1.2 Sensitivity and specificity

For a diagnostic test let X = true disease status and Y = diagnosis (positive/negative).

Definition (Sensitivity, specificity)

Sensitivity $= P(Y = \text{positive} \mid X = \text{diseased}) = \pi_{1|1}$; *specificity* $= P(Y = \text{negative} \mid X = \text{not diseased}) = \pi_{2|2}$.

For mammography combined with clinical breast examination, a literature survey estimated sensitivity 0.82 and specificity 0.99: 82% of women with breast cancer, and 99% of those without, are diagnosed correctly.

2.1.3 Independence of categorical variables

Definition (Independence)

X and Y are *independent* if $\pi_{ij} = \pi_{i+}\pi_{+j}$ for all i, j.

Proposition 2.2 (Independence as homogeneity)

If all $\pi_{i+} > 0$, then X and Y are independent if and only if $\pi_{j|1} = \pi_{j|2} = \cdots = \pi_{j|I}$ for every j; the common value is then π_{+j}.

Proof.

Step 1. If $\pi_{ij} = \pi_{i+}\pi_{+j}$ then $\pi_{j|i} = \pi_{ij}/\pi_{i+} = \pi_{+j}$ for every i.

Step 2. Conversely, if $\pi_{j|i} = c_j$ for all i, then $\pi_{+j} = \sum_i \pi_{i+}\pi_{j|i} = c_j \sum_i \pi_{i+} = c_j$, so $\pi_{ij} = \pi_{i+}\pi_{j|i} = \pi_{i+}\pi_{+j}$. ■

When Y is a response this *homogeneity of conditional distributions* is the more natural definition. Sample analogues: $p_{ij} = n_{ij}/n$, $p_{j|i} = n_{ij}/n_{i+}$, $n = \sum_i \sum_j n_{ij}$; estimates may also be written $\hat{\pi}_{ij}$.

2.1.4 Poisson, binomial, and multinomial sampling

(a) *Poisson sampling*: the IJ counts Y_{ij} are independent Poisson(μ_{ij}), with joint pmf $\prod_i \prod_j e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!$; nothing is fixed.

(b) *Multinomial sampling*: n is fixed and $\{n_{ij}\}$ is multinomial over the IJ cells, $\frac{n!}{\prod\prod n_{ij}!} \prod\prod \pi_{ij}^{n_{ij}}$.

(c) *Independent (product) multinomial sampling*: the row totals $n_i = n_{i+}$ are fixed and each row is an independent multinomial sample with probabilities $\{\pi_{j|i}\}$:

$$\prod_i \Big[\frac{n_i!}{\prod_j n_{ij}!} \prod_j \pi_{j|i}^{n_{ij}} \Big]. \tag{2.1}$$

(d) When both margins are fixed the distribution is hypergeometric (Section 3.5).

Theorem 2.3 (Conditioning on row totals gives product multinomial)

If $\{n_{ij}\}$ arise from independent Poisson sampling with means $\{\mu_{ij}\}$, or from multinomial sampling with probabilities $\pi_{ij} = \mu_{ij}/n$, then conditional on the row totals $\{n_{i+}\}$ the rows are independent multinomial samples with probabilities $\pi_{j|i} = \mu_{ij}/\mu_{i+}$.

Proof. [Supplement]

Step 1 (Poisson). The rows are independent sets of independent Poisson counts. By Theorem 1.5 applied to row i, given n_{i+} the counts $(n_{i1}, \ldots, n_{iJ})$ are multinomial$(n_{i+}, \{\mu_{ij}/\mu_{i+}\})$. Since row i's counts and total depend only on row i's Poisson variables, conditioning on all row totals keeps the rows independent.

Step 2 (multinomial). The row totals $(n_{1+}, \ldots, n_{I+})$ are multinomial$(n, \{\pi_{i+}\})$ (collapsing categories of a multinomial). Hence for tables with these row totals,

$$P(\{n_{ij}\} \mid \{n_{i+}\}) = \frac{\frac{n!}{\prod_i \prod_j n_{ij}!} \prod_i \prod_j \pi_{ij}^{n_{ij}}}{\frac{n!}{\prod_i n_{i+}!} \prod_i \pi_{i+}^{n_{i+}}} = \prod_i \Big[\frac{n_{i+}!}{\prod_j n_{ij}!} \prod_j \Big(\frac{\pi_{ij}}{\pi_{i+}} \Big)^{n_{ij}} \Big],$$

using $\pi_{i+}^{n_{i+}} = \prod_j \pi_{i+}^{n_{ij}}$. This is (2.1) with $\pi_{j|i} = \pi_{ij}/\pi_{i+} = \mu_{ij}/\mu_{i+}$. ■

So when X is explanatory it is sensible to analyse conditional on the row totals, as if they formed separate independent samples, even when the design did not fix them.

2.1.5 Seat belt example

Example 2.4 (Seat belts and crash outcome)

A study of seat-belt use (yes/no) by crash outcome (fatal/nonfatal) on the Massachusetts Turnpike illustrates the sampling schemes:
(a) recording all accidents in the coming year: n random, the four counts independent Poisson;
(b) sampling 200 police records from last year: $n = 200$ fixed, multinomial over four cells;
(c) sampling 100 fatal and 100 nonfatal records: column totals fixed, each column an independent binomial;
(d) the (unethical) experiment randomizing 100 of 200 subjects to wear belts and forcing crashes: row totals fixed, independent binomial rows.

2.1.6 Types of studies

Example 2.5 (Smoking and lung cancer: a case–control study)

Doll and Hill queried 709 lung cancer patients in 20 London hospitals about smoking, and for each a matched noncancer control (same hospital, gender, 5-year age group).

	Lung cancer	
Smoker	Cases	Controls
Yes	688	650
No	21	59
Total	709	709

The design fixes the lung cancer margin; we can estimate $P(\text{smoker} \mid \text{case}) = 688/709 = 0.970$ and $P(\text{smoker} \mid \text{control}) = 650/709 = 0.917$, but not $P(\text{cancer} \mid \text{smoking})$ without knowing the population prevalence (Bayes' theorem).

Definition (Study designs)

(a) *Retrospective* (case–control): samples are taken at fixed levels of the outcome Y and the explanatory X is looked up in the past; each column is a multinomial sample on X.
(b) *Prospective*: groups defined by X are followed into the future; *clinical trials* randomize subjects to X, *cohort studies* let subjects choose. Each row is a multinomial sample on Y.
(c) *Cross-sectional*: subjects are sampled and classified on both variables at once; the IJ counts are one multinomial sample.

Case–control, cohort and cross-sectional studies are *observational*; a clinical trial is *experimental*, and randomization balances the groups on other variables. Observational studies are common but more prone to bias.

2.2 Comparing two proportions

Motivation. With two groups and a binary response the association is a comparison of two success probabilities π_1, π_2. Three summaries are used—difference, ratio, and odds ratio—and they behave very differently under change of sampling design.

In a 2×2 table write $\pi_i = \pi_{1|i}$ for the success probability in row i, so $\pi_{2|i} = 1 - \pi_i$.

2.2.1 Difference of proportions

The *difference of proportions* $\pi_1 - \pi_2 \in [-1, 1]$ is the same whether we compare successes or failures, since $(1-\pi_1) - (1-\pi_2) = -(\pi_1 - \pi_2)$. It is 0 iff the rows have identical conditional distributions, i.e. iff Y is independent of the row classification. The difference comparing columns is in general different.

2.2.2 Relative risk

A difference of fixed size may matter more near 0 or 1: $0.010 - 0.001$ and $0.410 - 0.401$ are both 0.009, but the ratios are 10.0 and 1.02.

Definition (Relative risk)

The *relative risk* is $\pi_1/\pi_2 \in [0, \infty)$; it equals 1 under independence. Comparing the second response category gives a different value $(1-\pi_1)/(1-\pi_2)$.

2.2.3 Odds ratio

Definition (Odds, odds ratio)

For success probability π the *odds* are $\Omega = \pi/(1-\pi)$, inverted by $\pi = \Omega/(\Omega+1)$. With $\Omega_i = \pi_i/(1-\pi_i) = \pi_{i1}/\pi_{i2}$, the *odds ratio* is

$$\theta = \frac{\Omega_1}{\Omega_2} = \frac{\pi_1/(1-\pi_1)}{\pi_2/(1-\pi_2)} = \frac{\pi_{11}\pi_{22}}{\pi_{12}\pi_{21}}, \tag{2.2}$$

also called the *cross-product ratio* (Yule). The sample odds ratio is $\hat{\theta} = n_{11}n_{22}/(n_{12}n_{21})$.

E.g. $\pi = 0.75$ gives $\Omega = 3$: a success is three times as likely as a failure; $\Omega = 1/3$ gives $\pi = 0.25$.

2.2.4 Properties of the odds ratio

Proposition 2.6 (Properties of the odds ratio)

Assume all $\pi_{ij} > 0$.
(a) $\theta = 1$ iff $\Omega_1 = \Omega_2$ iff $\pi_1 = \pi_2$ (independence); $\theta > 1$ iff $\pi_1 > \pi_2$.
(b) Reversing the order of the rows (or of the columns) replaces θ by $1/\theta$, i.e. $\log\theta$ by $-\log\theta$; transposing the table leaves θ unchanged.
(c) θ is the same whichever conditional direction is used:

$$\theta = \frac{P(Y=1 \mid X=1)/P(Y=2 \mid X=1)}{P(Y=1 \mid X=2)/P(Y=2 \mid X=2)} = \frac{P(X=1 \mid Y=1)/P(X=2 \mid Y=1)}{P(X=1 \mid Y=2)/P(X=2 \mid Y=2)}. \tag{2.3}$$

(d) $\hat{\theta}$ is unchanged when both counts in a row, or both counts in a column, are multiplied by a positive constant.

Proof.

Step (a). $x \mapsto x/(1-x)$ is strictly increasing on $(0,1)$, so $\Omega_1 \gtreqless \Omega_2$ iff $\pi_1 \gtreqless \pi_2$.

Step (b). Swapping rows maps $\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})$ to $\pi_{21}\pi_{12}/(\pi_{22}\pi_{11}) = 1/\theta$; similarly for columns. Transposing swaps π_{12} and π_{21}, which leaves the cross-product ratio unchanged.

Step (c). The first ratio is $(\pi_{11}/\pi_{12})/(\pi_{21}/\pi_{22})$ after cancelling π_{1+} and π_{2+}; the second is $(\pi_{11}/\pi_{21})/(\pi_{12}/\pi_{22})$ after cancelling π_{+1}, π_{+2}. Both equal $\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})$.

Step (d). Multiplying row 1 by c multiplies numerator $n_{11}n_{22}$ and denominator $n_{12}n_{21}$ each by c; columns likewise. ■

Thus $\theta = 4$ means the odds (not the probability) in row 1 are four times those in row 2, and $\log 4 = 1.39$ and $\log 0.25 = -1.39$ represent equally strong associations. By (c) and (d), θ is equally valid for prospective, retrospective and cross-sectional designs: $\hat{\theta}$ estimates the same parameter whether a flu–vaccination case–control study samples $100 + 100$ or $40 + 160$ subjects. The sample difference and relative risk are invariant to scaling within rows, but not within columns or under transposition.

2.2.5 Aspirin and heart attacks revisited

Example 2.7 (Three measures for the aspirin study)

Combining fatal and nonfatal attacks in Example 2.1: placebo 189/11,034 = 0.0171, aspirin 104/11,037 = 0.0094.
(a) Difference: $0.0171 - 0.0094 = 0.0077$.
(b) Relative risk: $0.0171/0.0094 = 1.82$.
(c) Odds ratio: $\hat{\theta} = \dfrac{189 \times 10{,}933}{10{,}845 \times 104} = 1.83$; the odds of heart attack were 83% higher with placebo.

2.2.6 Case–control studies and the odds ratio

In a case–control study we can estimate $P(X = i \mid Y = j)$ but usually not $P(Y = j \mid X = i)$, hence neither the difference nor the relative risk of the outcome. By (2.3), however, the odds ratio is determined by conditional probabilities in either direction.

Example 2.8 (Odds ratio from Doll and Hill's data)

From Example 2.5,

$$\hat{\theta} = \frac{(688/709)/(21/709)}{(650/709)/(59/709)} = \frac{688 \times 59}{650 \times 21} = 3.0.$$

Using the direction of interest: the estimated odds of lung cancer for smokers were 3.0 times those for nonsmokers.

2.2.7 Relationship between odds ratio and relative risk

Proposition 2.9 (Odds ratio versus relative risk)

$$\theta = \frac{\pi_1}{\pi_2} \times \frac{1-\pi_2}{1-\pi_1}.$$

Hence $\theta \approx$ relative risk when π_1 and π_2 are both close to 0, and θ is farther from 1 than the relative risk (on the same side).

Proof.

Step 1. Directly from (2.2): $\theta = \frac{\pi_1}{1-\pi_1} \cdot \frac{1-\pi_2}{\pi_2}$.

Step 2. If $\pi_1 > \pi_2$ then $(1-\pi_2)/(1-\pi_1) > 1$ so $\theta > \pi_1/\pi_2 > 1$; if $\pi_1 < \pi_2$ the factor is < 1 and $\theta < \pi_1/\pi_2 < 1$. As $\pi_1, \pi_2 \to 0$ the factor tends to 1. ■

In the aspirin study ($\pi_i < 0.02$) the relative risk 1.82 and odds ratio 1.83 almost agree. Hence in case–control studies of rare diseases $\hat{\theta}$ roughly estimates the unestimable relative risk (Cornfield, 1951): smokers had roughly 3 times the lung cancer rate of nonsmokers.

2.3 Partial association in stratified 2×2 tables

Motivation. In observational studies an apparent effect of X on Y may reflect a third variable Z that affects both (confounding). Controlling for Z by holding it fixed can change the association dramatically, even reversing its direction.

To study a passive-smoking effect on lung cancer, for example, one must control age and socioeconomic status: spouses of nonsmokers may be younger, and younger people get lung cancer less often.

2.3.1 Partial tables

Definition (Partial and marginal tables)

Fixing the levels of Z, the two-way cross-sections of an $X \times Y \times Z$ table are the *partial tables*; their associations are *conditional associations*. Summing the partial tables gives the XY *marginal table*, which ignores Z rather than controlling it.

2.3.2 Death penalty example

Example 2.10 (Death penalty verdict by defendant's and victims' race)

674 defendants in Florida multiple-murder indictments, 1976–1987. Y = death penalty, X = defendant's race, Z = victims' race.

Victims' race	Defendant's race	Yes	No	Percent yes
White	White	53	414	11.3
	Black	11	37	22.9
Black	White	0	16	0.0
	Black	4	139	2.8
Total	White	53	430	11.0
	Black	15	176	7.9

(a) Controlling for victims' race, black defendants received the death penalty more often: by $22.9 - 11.3 = 11.6$ points with white victims and 2.8 points with black victims.
(b) Ignoring victims' race, white defendants received it more often (11.0% vs. 7.9%): the association reverses.
(c) Explanation: victims' race is strongly associated with defendant's race (marginal odds ratio $467 \times 143/(48 \times 16) = 87.0$), and the death penalty is much more likely when the victims are white. Whites mostly kill whites, and killing whites more often brings the death penalty.

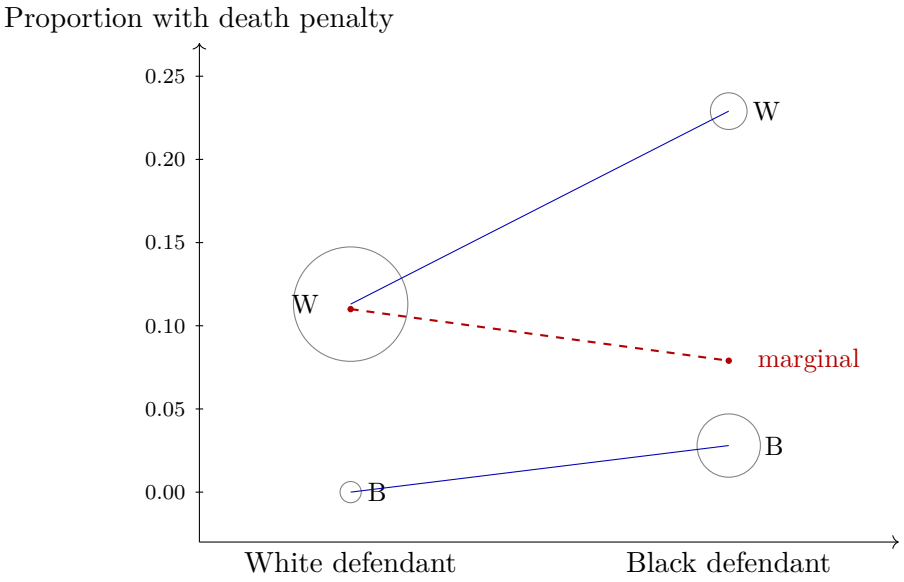


Figure 1: Simpson's paradox in the death penalty data. Letters give victims' race; circle area is proportional to the number of cases. Both conditional lines rise, the marginal line falls, since the summary proportions lie near the large circles.

Definition (Simpson's paradox)

A marginal association whose direction differs from that of each conditional association is an instance of *Simpson's paradox* (Yule 1903, Simpson 1951).

It warns against causal interpretation of an X–Y association. Fisher argued that some factor (e.g. genetic) might explain away the smoking–lung cancer association; Cornfield showed that for a very strong association, the confounder would itself need a very strong association with both X and Y.

2.3.3 Conditional and marginal odds ratios

For a $2 \times 2 \times K$ table with expected frequencies μ_{ijk}:

$$\theta_{XY(k)} = \frac{\mu_{11k}\mu_{22k}}{\mu_{12k}\mu_{21k}} \quad \text{(conditional)}, \qquad \theta_{XY} = \frac{\mu_{11+}\mu_{22+}}{\mu_{12+}\mu_{21+}} \quad \text{(marginal)},$$

with $\mu_{ij+} = \sum_k \mu_{ijk}$; sample versions use counts.

Example 2.11 (Odds ratios for the death penalty data)

(a) White victims: $\hat{\theta}_{XY(1)} = \dfrac{53 \times 37}{414 \times 11} = 0.43$; the odds of the death penalty for white defendants were 43% of those for black defendants.
(b) Black victims: $\hat{\theta}_{XY(2)} = \dfrac{0 \times 139}{16 \times 4} = 0.0$ (no white defendant with black victims got the death penalty).
(c) Marginal: $\hat{\theta}_{XY} = \dfrac{53 \times 176}{430 \times 15} = 1.45$; the odds were 45% higher for white defendants.

2.3.4 Marginal versus conditional independence

Definition (Conditional independence)

X and Y are *conditionally independent at level* k of Z if they are independent in partial table k, i.e. $P(Y = j \mid X = i, Z = k) = P(Y = j \mid Z = k)$ for all i, j. They are *conditionally independent given* Z if this holds for every k.

Proposition 2.12 (Factorization under conditional independence)

For a single multinomial over the three-way table with $\pi_{ijk} = P(X = i, Y = j, Z = k)$ and all $\pi_{++k} > 0$, X and Y are conditionally independent given Z iff

$$\pi_{ijk} = \frac{\pi_{i+k}\,\pi_{+jk}}{\pi_{++k}} \qquad \text{for all } i, j, k.$$

Conditional independence does not imply marginal independence: in general $\pi_{ij+} = \sum_k \pi_{i+k}\pi_{+jk}/\pi_{++k} \neq \pi_{i++}\pi_{+j+}$.

Proof.

Step 1. Always $\pi_{ijk} = P(X = i, Z = k)\,P(Y = j \mid X = i, Z = k) = \pi_{i+k}\,P(Y = j \mid X = i, Z = k)$.

Step 2. Under conditional independence the last factor equals $P(Y = j \mid Z = k) = \pi_{+jk}/\pi_{++k}$, giving the factorization. Conversely, the factorization gives $P(Y = j \mid X = i, Z = k) = \pi_{ijk}/\pi_{i+k} = \pi_{+jk}/\pi_{++k}$ whenever $\pi_{i+k} > 0$.

Step 3. Summing over k gives $\pi_{ij+} = \sum_k \pi_{i+k}\pi_{+jk}/\pi_{++k}$, which involves k in every factor and does not reduce to $\pi_{i++}\pi_{+j+}$; Example 2.13 is an explicit counterexample. ■

For $2 \times 2 \times K$ tables, conditional independence means $\theta_{XY(k)} = 1$ for all k.

Example 2.13 (Conditional but not marginal independence)

Expected frequencies for Y = response, X = treatment, Z = clinic:

Clinic	Treatment	Success	Failure
1	A	18	12
	B	12	8
2	A	2	8
	B	8	32
Total	A	20	20
	B	20	40

(a) $\theta_{XY(1)} = \dfrac{18 \times 8}{12 \times 12} = 1.0$ and $\theta_{XY(2)} = \dfrac{2 \times 32}{8 \times 8} = 1.0$: conditional independence.

(b) Marginal: $\theta_{XY} = \dfrac{20 \times 40}{20 \times 20} = 2.0$.

(c) The XZ and YZ conditional odds ratios all equal 6.0, e.g. $\dfrac{18 \times 8}{12 \times 2} = 6.0$: clinic 1 uses treatment A more and also has more successes (perhaps its patients are healthier). Concluding from the marginal table that A is better would be misleading.

2.3.5 Homogeneous association

Definition (Homogeneous association)

A $2 \times 2 \times K$ table has *homogeneous XY association* if $\theta_{XY(1)} = \theta_{XY(2)} = \cdots = \theta_{XY(K)}$. Conditional independence is the special case where the common value is 1. If the conditional odds ratios vary with k there is *interaction*, and Z is an *effect modifier*.

Proposition 2.14 (Homogeneous association is symmetric)

For a $2 \times 2 \times 2$ table with positive probabilities,

$$\frac{\theta_{XY(1)}}{\theta_{XY(2)}} = \frac{\theta_{XZ(1)}}{\theta_{XZ(2)}} = \frac{\theta_{YZ(1)}}{\theta_{YZ(2)}},$$

where $\theta_{XZ(j)}$ is the XZ odds ratio at level j of Y and $\theta_{YZ(i)}$ the YZ odds ratio at level i of X. Hence, in a $2 \times 2 \times K$ table, homogeneous XY association holds iff for every pair of levels k, k' of Z the XZ (and the YZ) odds ratio formed with k, k' is the same at both levels of Y (of X).

Proof. [Supplement]

Step 1. Directly,

$$\frac{\theta_{XY(1)}}{\theta_{XY(2)}} = \frac{\pi_{111}\pi_{221}}{\pi_{121}\pi_{211}} \cdot \frac{\pi_{122}\pi_{212}}{\pi_{112}\pi_{222}}, \qquad \frac{\theta_{XZ(1)}}{\theta_{XZ(2)}} = \frac{\pi_{111}\pi_{212}}{\pi_{112}\pi_{211}} \cdot \frac{\pi_{122}\pi_{221}}{\pi_{121}\pi_{222}}.$$

Both equal $\dfrac{\pi_{111}\pi_{122}\pi_{212}\pi_{221}}{\pi_{112}\pi_{121}\pi_{211}\pi_{222}}$ (numerator: the cells whose index sum is odd; denominator: those whose index sum is even). The same expression arises for $\theta_{YZ(1)}/\theta_{YZ(2)}$ by the symmetry of this formula in the three indices.

Step 2. For $2 \times 2 \times K$, apply Step 1 to the $2 \times 2 \times 2$ subtable formed by levels k, k' of Z: $\theta_{XY(k)} = \theta_{XY(k')}$ iff the XZ odds ratio for (k, k') is the same at $Y = 1$ and $Y = 2$, and likewise for YZ. ■

So homogeneous association is a property of the three variables jointly ("no three-factor interaction"). For instance, if for smoking and lung cancer the odds ratios were 1.2, 3.9, 8.8 in age groups

< 45, 45–65, > 65, age would be an effect modifier. For the death penalty data $\hat{\theta}_{XY(1)} = 0.43$ and $\hat{\theta}_{XY(2)} = 0.0$; the latter is unstable (zero count), and adding 1/2 to each cell gives 0.94, so the data need not contradict homogeneity (tests in Section 6.3).

2.4 Extensions for $I \times J$ tables

Motivation. Beyond 2×2 tables a single number rarely captures the association. A minimal set of odds ratios describes it completely, while summary indices (entropy-based for nominal, concordance-based for ordinal variables) trade completeness for interpretability.

2.4.1 Odds ratios in $I \times J$ tables

Any pair of rows a, b and columns c, d gives an odds ratio $\pi_{ac}\pi_{bd}/(\pi_{bc}\pi_{ad})$; there are $\binom{I}{2}\binom{J}{2}$ of them, highly redundant.

Definition (Local and baseline odds ratios)

The *local odds ratios* are

$$\theta_{ij} = \frac{\pi_{ij}\,\pi_{i+1,j+1}}{\pi_{i,j+1}\,\pi_{i+1,j}}, \qquad i = 1, \ldots, I-1,\ j = 1, \ldots, J-1. \tag{2.4}$$

An alternative basic set uses the last row and column as baseline:

$$\alpha_{ij} = \frac{\pi_{ij}\,\pi_{IJ}}{\pi_{Ij}\,\pi_{iJ}}, \qquad i = 1, \ldots, I-1,\ j = 1, \ldots, J-1. \tag{2.5}$$

Proposition 2.15 ($(I-1)(J-1)$ odds ratios describe the association)

Assume all $\pi_{ij} > 0$.
(a) Every odds ratio $\pi_{ac}\pi_{bd}/(\pi_{bc}\pi_{ad})$ with $a < b$, $c < d$ is the product $\prod_{i=a}^{b-1}\prod_{j=c}^{d-1}\theta_{ij}$.
(b) Given the marginal distributions $\{\pi_{i+}\}, \{\pi_{+j}\}$, the local odds ratios determine the cell probabilities uniquely.
(c) Independence holds iff all $(I-1)(J-1)$ local odds ratios equal 1.

Proof. [Supplement]

Step (a). Let $\lambda_{ij} = \log \pi_{ij}$ and $\log \theta_{ij} = \lambda_{ij} - \lambda_{i,j+1} - \lambda_{i+1,j} + \lambda_{i+1,j+1}$. Summing over $j = c, \ldots, d-1$ telescopes to $\lambda_{ic} - \lambda_{id} - \lambda_{i+1,c} + \lambda_{i+1,d}$; summing that over $i = a, \ldots, b-1$ telescopes to $\lambda_{ac} - \lambda_{ad} - \lambda_{bc} + \lambda_{bd}$, the log of the stated odds ratio.

Step (b). Suppose $\{\pi_{ij}\}$ and $\{\pi'_{ij}\}$ have the same margins and the same local odds ratios. Let $\delta_{ij} = \log(\pi'_{ij}/\pi_{ij})$. Its local contrasts $\delta_{ij} - \delta_{i,j+1} - \delta_{i+1,j} + \delta_{i+1,j+1}$ vanish, so by the telescoping in (a) so does $\delta_{ij} - \delta_{iJ} - \delta_{Ij} + \delta_{IJ}$. With $r_i = \delta_{iJ} - \delta_{IJ}$ and $c_j = \delta_{Ij}$ this gives $\delta_{ij} = r_i + c_j$. Hence

$$\sum_i \sum_j (\pi'_{ij} - \pi_{ij}) \log \frac{\pi'_{ij}}{\pi_{ij}} = \sum_i r_i(\pi'_{i+} - \pi_{i+}) + \sum_j c_j(\pi'_{+j} - \pi_{+j}) = 0.$$

Each term $(\pi' - \pi)\log(\pi'/\pi)$ is ≥ 0, with equality iff $\pi' = \pi$ (log is increasing). Hence $\pi'_{ij} = \pi_{ij}$ for all i, j.

Step (c). If $\pi_{ij} = \pi_{i+}\pi_{+j}$ then every $\theta_{ij} = 1$. Conversely, the independence table $\{\pi_{i+}\pi_{+j}\}$ has the same margins and all local odds ratios 1; by (b) it is the only such table. ∎

In this sense $(I-1)(J-1)$ parameters describe any association in an $I \times J$ table. For Example 2.1 the sample local odds ratios are $\frac{18\times 99}{171\times 5} = 2.08$ (fatal vs. nonfatal) and $\frac{171\times 10{,}933}{99\times 10{,}845} = 1.74$ (nonfatal vs. none); their product 3.63 is the odds ratio for the first and third columns. In each case the more serious outcome was more prevalent under placebo. For $I \times J \times K$ tables, homogeneous XY association means every conditional XY odds ratio is the same at each level of Z.

2.4.2 Summary measures of association

A *proportional reduction in variation* measure, with $V(Y)$ a variation measure for the marginal distribution $\{\pi_{+j}\}$ and $V(Y \mid i)$ that for $\{\pi_{j|i}\}$, is

$$\frac{V(Y) - E[V(Y \mid X)]}{V(Y)}, \qquad E[V(Y \mid X)] = \sum_i \pi_{i+} V(Y \mid i), \tag{2.6}$$

generalizing R^2. For a nominal response Theil used the *entropy* $V(Y) = -\sum_j \pi_{+j} \log \pi_{+j}$, giving the *uncertainty coefficient*

$$U = -\frac{\sum_i \sum_j \pi_{ij} \log(\pi_{ij}/(\pi_{i+}\pi_{+j}))}{\sum_j \pi_{+j} \log \pi_{+j}}. \tag{2.7}$$

Proposition 2.16 (Range of the uncertainty coefficient)

If more than one $\pi_{+j} > 0$, then $0 \le U \le 1$; $U = 0$ iff X and Y are independent, and $U = 1$ iff for each i some $\pi_{j|i} = 1$.

Proof. [Supplement] Write

$$H(Y) = -\sum_j \pi_{+j} \log \pi_{+j} > 0, \qquad H(Y \mid X) = -\sum_i \pi_{i+} \sum_j \pi_{j|i} \log \pi_{j|i} \ge 0.$$

Step 1. (2.7) is (2.6) with entropy: since $\log \pi_{j|i} = \log \pi_{ij} - \log \pi_{i+}$, $H(Y) - H(Y \mid X) = \sum_i \sum_j \pi_{ij} \log \frac{\pi_{ij}}{\pi_{i+}\pi_{+j}}$, so $U = [H(Y) - H(Y \mid X)]/H(Y)$.

Step 2. By Jensen, $-\sum\sum \pi_{ij} \log \frac{\pi_{i+}\pi_{+j}}{\pi_{ij}} \ge -\log \sum\sum \pi_{i+}\pi_{+j} \ge 0$, with equality iff $\pi_{ij} = \pi_{i+}\pi_{+j}$ on all cells (then the sum is 1). So $U \ge 0$, with equality iff independence.

Step 3. $U \le 1$ iff $H(Y \mid X) \ge 0$, true since each $-\pi \log \pi \ge 0$; equality iff every conditional distribution is degenerate. ■

It is hard to develop intuition for such measures (what is a "30% reduction in entropy"?); summaries are easier to interpret when both variables are ordinal.

2.4.3 Ordinal trends: concordant and discordant pairs

For ordinal X and Y a monotone trend is common. A pair of subjects is *concordant* if the subject ranking higher on X also ranks higher on Y, *discordant* if it ranks lower on Y, and *tied* if they agree on X and/or Y.

Example 2.17 (Job satisfaction and income)

Black males in the 1996 General Social Survey ($n = 96$); VD = very dissatisfied, LD = a little dissatisfied, MS = moderately satisfied, VS = very satisfied.

Income ($)	VD	LD	MS	VS
$< 15{,}000$	1	3	10	6
15,000–25,000	2	3	10	7
25,000–40,000	1	6	14	12
$> 40{,}000$	0	1	9	11

Each cell's count times the total count below and to the right gives concordant pairs:

$$\begin{aligned} C = {} & 1(3+10+7+6+14+12+1+9+11) + 3(10+7+14+12+9+11) \\ & + 10(7+12+11) + 2(6+14+12+1+9+11) + 3(14+12+9+11) + 10(12+11) \\ & + 1(1+9+11) + 6(9+11) + 14(11) = 1331, \end{aligned}$$

and similarly (below and to the left) $D = 3(2{+}1{+}0){+}10(2{+}3{+}1{+}6{+}0{+}1){+}\cdots{+}12(0{+}1{+}9) = 849$. Since $C > D$, low income tends to occur with low satisfaction.

For two independent observations from $\{\pi_{ij}\}$ the probabilities of concordance and discordance are

$$\Pi_c = 2\sum_i \sum_j \pi_{ij}\Big(\sum_{h>i}\sum_{k>j} \pi_{hk}\Big), \qquad \Pi_d = 2\sum_i \sum_j \pi_{ij}\Big(\sum_{h>i}\sum_{k<j} \pi_{hk}\Big), \tag{2.8}$$

the factor 2 because either observation may be the one in cell (i, j).

2.4.4 Ordinal measure of association: gamma

Definition (Goodman–Kruskal gamma)

Given that a pair is untied on both variables, $\Pi_c/(\Pi_c + \Pi_d)$ is the probability of concordance. *Gamma* is

$$\gamma = \frac{\Pi_c - \Pi_d}{\Pi_c + \Pi_d}, \qquad \text{sample version } \hat{\gamma} = \frac{C - D}{C + D}. \tag{2.9}$$

Proposition 2.18 (Properties of gamma)

(a) $-1 \le \gamma \le 1$; $\gamma = 1$ iff $\Pi_d = 0$ and $\gamma = -1$ iff $\Pi_c = 0$ (perfect monotonicity, not linearity).
(b) Reversing the category order of one variable changes the sign of γ; γ is symmetric in X and Y.
(c) Independence implies $\gamma = 0$; the converse is false.

Proof. [Supplement]

Step (a). $\Pi_c, \Pi_d \ge 0$, so $|\Pi_c - \Pi_d| \le \Pi_c + \Pi_d$, with the extreme cases as stated.

Step (b). Reversing one ordering turns concordant pairs into discordant ones and vice versa, swapping Π_c and Π_d. Interchanging the roles of X and Y does not change whether a pair is concordant.

Step (c). Let (X_1, Y_1), (X_2, Y_2) be independent draws. Then $\Pi_c = 2P(X_1 > X_2, Y_1 > Y_2)$ and $\Pi_d = 2P(X_1 > X_2, Y_1 < Y_2)$. Under independence of X and Y, (X_1, X_2) is independent of (Y_1, Y_2), so $\Pi_c = 2P(X_1 > X_2)P(Y_1 > Y_2)$ and $\Pi_d = 2P(X_1 > X_2)P(Y_1 < Y_2)$; since Y_1, Y_2 are i.i.d., $P(Y_1 > Y_2) = P(Y_1 < Y_2)$, so $\Pi_c = \Pi_d$ and $\gamma = 0$. For the converse, a U-shaped joint distribution can have $\Pi_c = \Pi_d$ without independence, e.g. the 3×3 table with $\pi_{11} = \pi_{31} = \pi_{22} = 1/3$ has $\Pi_c = 2\pi_{11}\pi_{22} = 2/9 = 2\pi_{22}\pi_{31} = \Pi_d$. ∎

2.4.5 Gamma for job satisfaction example

Example 2.19 (Gamma for income and job satisfaction)

From Example 2.17, $\hat{\gamma} = (1331-849)/(1331+849) = 0.221$: of the untied pairs, the proportion concordant exceeds the proportion discordant by 0.221, a weak tendency for satisfaction to increase with income.